

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: CSA51 **COURSE NAME:** Cryptography and Network Security

COURSE OUTCOME COVERED

CO4: *Implement best security practices for IP, email, and web service using PGP, S/MIME for enhancing email Security.CO (BL3, BL6)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

TOTAL MARKS: 25

Annexure A

Industry Problem-Solving Task

Code of Conduct:

I Pusalapati Chandu(192372119) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Pchandu

Annexure A

1. SSL/TLS Overhead and Web Server Performance Optimization

A web server uses TLS 1.3 for secure communication. Each HTTP response carries 2000 bytes of payload. TLS adds 25 bytes (record header + authentication tag) and handshake overhead of 3 KB per session. The server handles 10,000 requests per second, with 1,000 new TLS handshakes per second.

Task:

- A. Calculate the total data transmitted per second including TLS overhead.
 - B. Determine the percentage overhead due to TLS encryption.
 - C. Estimate bandwidth consumed due to handshake overhead per second.
 - D. Analyze the trade-off between security and latency, and propose optimization techniques (e.g., session resumption, HTTP/2, or QUIC).
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2. PGP-Based Email Encryption Load in Enterprise System

An enterprise email server encrypts emails using PGP. Each email contains a 10 MB attachment. PGP introduces 800 bytes overhead for key encryption and signatures. The system processes 5,000 emails per day.

Task:

- A. Calculate the total data transmitted per day including overhead.
 - B. Determine the percentage increase in storage due to PGP.
 - C. Analyze the computational overhead during encryption/decryption.
 - D. Recommend optimization strategies (e.g., compression, batching, or hardware acceleration).
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3. IPSec Transport vs Tunnel Mode Performance Comparison

A company compares IPSec transport mode and tunnel mode for securing internal communications. Original packet size = 1000 bytes payload + 20-byte header. Transport mode adds 30 bytes, while tunnel mode adds 50 bytes overhead. Traffic volume = 20,000 packets per second.

Task:

- A. Compute total packet size for both modes.
 - B. Calculate bandwidth usage for each mode per second.
 - C. Compare efficiency and security benefits of both modes.
 - D. Recommend the appropriate mode for internal vs external communication with justification.
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4. S/MIME Certificate Management in Large Organization

An organization manages 12,000 employees using S/MIME certificates. Each certificate

is 2 KB, and certificate renewal happens every year. During renewal, both old and new certificates are stored simultaneously for 1 month.

Task:

- A. Calculate total storage required during the renewal period.
 - B. Estimate certificate distribution overhead across the network.
 - C. Analyze challenges in certificate lifecycle management at scale.
 - D. Propose optimization strategies (e.g., automation, centralized PKI, or short-lived certificates).
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5. Email Encryption Latency in Cloud-Based Systems

A cloud email service encrypts outgoing emails using AES (data) and RSA (key exchange). AES encryption takes 0.2 ms per MB, and RSA key exchange takes 5 ms per email. Each email has 8 MB attachment. The system processes 8,000 emails per hour.

Task:

- A. Calculate total encryption time per email.
- B. Estimate total processing time per hour.
- C. Analyze the performance bottleneck between symmetric and asymmetric operations.
- D. Recommend optimization strategies (e.g., ECC adoption, parallel processing, or hardware acceleration).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

QUESTION 1 – SSL/TLS Overhead and Web Server Performance Optimization

The question states that each HTTP response contains 2000 bytes of payload. TLS adds 25 bytes to each response. A TLS handshake adds 3 KB per new session. The server handles 10,000 requests per second and 1,000 new handshakes per second. □ filecite□turn1file0□L19-L29□

1. Problem Understanding

- The server is using TLS 1.3 to secure web communication.
- TLS provides confidentiality, integrity and authentication, but it introduces some additional data and processing.
- There are two types of overhead given in the question: per-response TLS overhead and per-session handshake overhead.
- The high number of requests means even a small overhead per request becomes significant.
- The high number of new handshakes also increases network traffic and latency.

2. Given Data

Parameter	Value
HTTP payload per response	2000 bytes
TLS overhead per response	25 bytes
Requests per second	10,000
Handshake overhead	3 KB/session
New handshakes per second	1,000

3. Part A – Total Data Per Second

1. Data for one response = 2000 bytes.
2. TLS overhead for one response = 25 bytes.
3. Total response size = 2000 + 25 = 2025 bytes.
4. Requests per second = 10,000.

$$\text{Total data/second} = 2025 \times 10,000 = 20,250,000 \text{ bytes/s}$$

Therefore, the server transmits 20,250,000 bytes every second for the HTTP responses including the stated TLS record overhead.

Using decimal units, this is 20.25 MB/s.

4. Part B – Percentage TLS Overhead

5. TLS overhead = 25 bytes.
6. Original payload = 2000 bytes.

$$\text{Percentage overhead} = (25 / 2000) \times 100 = 1.25\%$$

Therefore, TLS adds 1.25% overhead to each response payload based on the values in the question.

5. Meaning of the Result

- For every 2000 bytes of application data, 25 additional bytes are transmitted.
- The percentage is small for one response, but at 10,000 requests per second it becomes a measurable bandwidth cost.
- TLS overhead is the price paid for secure communication.

6. Part C – Handshake Bandwidth Per Second

7. One handshake adds 3 KB.
8. New handshakes per second = 1000.
9. Using 1 KB = 1024 bytes, 3 KB = 3072 bytes.

$$\text{Handshake bandwidth} = 3072 \times 1000 = 3,072,000 \text{ bytes/s}$$

Therefore, the handshake overhead is approximately 3.072 MB/s using decimal MB, or 24.576 Mb/s.

7. Part D – Security and Latency Trade-off

- TLS is necessary because it protects web traffic from eavesdropping and tampering.
- A TLS handshake requires cryptographic processing and network communication.
- If a new handshake is created too often, users may experience additional connection latency.
- The server also spends CPU resources processing handshakes.
- Therefore, the goal is not to remove TLS, but to reduce unnecessary TLS setup operations.

8. Optimization Techniques

Technique	Simple Explanation	Benefit
Session resumption	Reuse information from an earlier secure session.	Reduces full handshake cost.
Connection reuse	Keep secure connections open for multiple requests.	Reduces repeated setup.
HTTP/2	Allows multiple requests over fewer connections.	Improves connection utilization.
QUIC / HTTP/3	Modern transport using TLS 1.3.	Can reduce connection setup latency.
Hardware acceleration	Use optimized cryptographic CPU features.	Reduces CPU processing time.

9. Best Recommendation

The first recommendation is to reduce the number of full TLS handshakes. Session resumption and connection reuse are suitable because the server has 1,000 new handshakes every second. HTTP/2 can further reduce connection overhead by multiplexing many requests. QUIC/HTTP/3 is another option when the system is ready to adopt it.

10. Conclusion

TLS adds measurable bandwidth and processing overhead, but the security benefit is essential for web communication. The calculated response traffic is 20.25 MB/s and the handshake overhead is approximately 3.072 MB/s. The best approach is to retain TLS while reducing repeated handshakes and improving connection efficiency.

QUESTION 2 – PGP-Based Email Encryption Load in Enterprise System

The question states that each enterprise email contains a 10 MB attachment. PGP adds 800 bytes of overhead for key encryption and signatures. The system processes 5,000 emails per day.

1. Problem Understanding

- PGP is used to protect email content and attachments.
- The large part of each email is the 10 MB attachment.
- The 800-byte PGP overhead is related to cryptographic information such as key protection and signatures.
- Although 800 bytes is small compared with 10 MB, it becomes measurable when 5,000 emails are processed every day.
- The question asks about data size, storage increase, computational overhead and optimization.

2. Given Data

Parameter	Value
Attachment per email	10 MB
PGP overhead per email	800 bytes
Emails per day	5,000

3. Part A – Total Data Per Day

10. Assume 1 MB = 1,000,000 bytes.
11. 10 MB = 10,000,000 bytes.
12. PGP overhead = 800 bytes.
13. Total per email = 10,000,000 + 800 = 10,000,800 bytes.
14. Number of emails = 5,000.

$$\text{Total/day} = 10,000,800 \times 5,000 = 50,004,000,000 \text{ bytes}$$

Therefore, the total transmitted data is 50.004 GB per day using decimal units.

4. Part B – Percentage Increase in Storage

15. Original data/day = 10,000,000 × 5,000 = 50,000,000,000 bytes.
16. PGP overhead/day = 800 × 5,000 = 4,000,000 bytes.

$$\text{Increase} = (4,000,000 / 50,000,000,000) \times 100 = 0.008\%$$

Therefore, PGP increases the stated storage/data amount by only 0.008%.

5. What This Means

- The PGP overhead is very small compared with the 10 MB attachment.
- The main challenge is therefore not storage overhead.
- CPU processing and encryption/decryption time are more important areas for optimization.
- The system should protect the emails without creating unnecessary processing delays.

6. Part C – Computational Overhead

- PGP uses a hybrid cryptographic design.
- Symmetric encryption is suitable for the large attachment because it is fast.
- Public-key cryptography is used for protecting key material rather than encrypting the complete large attachment.
- Digital signatures add processing but provide authentication and integrity.
- Decryption also requires cryptographic processing on the receiving side.
- At 5,000 emails per day, these operations can create a significant CPU workload even though the storage overhead is small.

7. Computational Work

Operation	Purpose	Impact
Symmetric encryption	Encrypt the 10 MB attachment.	Efficient for large data.
Key encryption	Protect the session/key information.	More computationally expensive, but small data.
Digital signature	Authenticate sender and protect integrity.	Adds processing.
Decryption	Recover email/attachment at receiver.	Consumes CPU resources.

8. Part D – Optimization Strategies

Strategy	How it helps
Compression	Reduces attachment size when the content is compressible.
Batching	Groups processing jobs to improve server utilization.
Parallel processing	Encrypts multiple independent emails at the same time.
Hardware acceleration	Speeds up cryptographic operations.
Efficient job scheduling	Prevents CPU overload during peak periods.

9. Recommended Solution

The organization should first use efficient symmetric encryption for large attachments, then optimize the public-key and signature operations. Parallel processing and hardware acceleration are useful when many emails are processed. Compression should be performed before encryption when the attachment type benefits from compression.

10. Conclusion

The total daily data including PGP overhead is 50.004 GB, while the percentage increase is only 0.008%. This shows that PGP has very little storage overhead for a 10 MB attachment. The main enterprise concern should be computational efficiency, so parallel processing, batching and hardware acceleration are suitable solutions.

QUESTION 3 – IPSec Transport vs Tunnel Mode Performance Comparison

The question compares IPSec transport and tunnel modes. The original packet contains 1000 bytes of payload and a 20-byte header. Transport mode adds 30 bytes and tunnel mode adds 50 bytes. Traffic volume is 20,000 packets per second. □ filecite □ turn1 file0 □ L40-L50 □

1. Problem Understanding

- IPSec provides security for IP communication.
- Transport mode and tunnel mode protect traffic differently.
- Transport mode normally keeps the original IP header visible and adds security protection to the payload.
- Tunnel mode encapsulates the original packet inside a new packet.
- Tunnel mode normally has more overhead, but it provides stronger encapsulation for VPN-style communication.

2. Given Data

Parameter	Value
Payload	1000 bytes
Original IP header	20 bytes
Transport overhead	30 bytes
Tunnel overhead	50 bytes
Traffic	20,000 packets/second

3. Part A – Packet Size

17. Original packet = payload + header = $1000 + 20 = 1020$ bytes.

18. Transport packet = $1020 + 30 = 1050$ bytes.

19. Tunnel packet = $1020 + 50 = 1070$ bytes.

Mode	Calculation	Final Size
Original	$1000 + 20$	1020 bytes
Transport	$1020 + 30$	1050 bytes
Tunnel	$1020 + 50$	1070 bytes

4. Part B – Bandwidth

20. Transport bandwidth = $1050 \times 20,000 = 21,000,000$ bytes/s.

21. Tunnel bandwidth = $1070 \times 20,000 = 21,400,000$ bytes/s.

Transport = 21 MB/s; Tunnel = 21.4 MB/s

Tunnel mode uses 400,000 bytes/s more bandwidth than transport mode under the supplied values.

5. Simple Comparison

- Transport mode is smaller and therefore more bandwidth-efficient.
- Tunnel mode has more overhead because the original packet is encapsulated.
- The difference is only 20 bytes per packet, but it becomes significant at 20,000 packets per second.

6. Part C – Efficiency and Security

Feature	Transport Mode	Tunnel Mode
Packet size	1050 bytes	1070 bytes
Bandwidth efficiency	Higher	Lower
Original IP header	Visible	Protected inside inner packet
Typical use	Host-to-host/internal	VPN and gateway communication
Encapsulation	Less	More

7. Transport Mode – Advantages

- Lower packet overhead.
- Lower bandwidth consumption.
- Suitable when endpoints themselves need IPSec protection.
- Useful for suitable internal host-to-host communication.

8. Tunnel Mode – Advantages

- Protects the complete original IP packet inside the tunnel.
- Useful for VPN communication.
- Can hide internal addressing information from the outside packet.
- Suitable for communication across an untrusted external network.

9. Part D – Recommendation

For internal host-to-host communication, transport mode is generally recommended when its security boundary is sufficient because it has lower overhead. For external communication across an untrusted network, tunnel mode is recommended because it encapsulates and protects the original packet.

10. Justification

22. Transport mode = 21 MB/s.

23. Tunnel mode = 21.4 MB/s.

24. Difference = 0.4 MB/s.

The numerical difference is small, but security requirements are more important than bandwidth alone. Therefore, internal communication can use transport mode where appropriate, while external VPN communication should prefer tunnel mode.

11. Conclusion

Transport mode provides better bandwidth efficiency, while tunnel mode provides stronger packet encapsulation. The correct choice depends on the communication environment. Security boundary and network exposure should be considered together with performance.

QUESTION 4 – S/MIME Certificate Management in Large Organization

The organization has 12,000 employees. Each S/MIME certificate is 2 KB. Renewal occurs every year, and during renewal both the old and new certificates are stored for one month.

□filecite□turn1file0□L51-L61□

1. Problem Understanding

- S/MIME certificates are used to support secure email operations.
- Every employee has a certificate.
- The organization must manage certificates throughout their lifecycle.
- During renewal, two certificates exist at the same time.
- At 12,000 employees, manual management becomes difficult.

2. Given Data

Parameter	Value
Employees	12,000
Certificate size	2 KB
Certificates during renewal	2 per employee
Overlap period	1 month
Renewal	Every year

3. Part A – Storage During Renewal

25. One certificate = 2 KB.

26. Old + new certificate = $2 + 2 = 4$ KB per employee.

27. Employees = 12,000.

$$\text{Total} = 12,000 \times 4 \text{ KB} = 48,000 \text{ KB}$$

28. $48,000 / 1024 = 46.875$ MiB.

Therefore, approximately 46.875 MiB of certificate storage is needed during the renewal overlap. Using decimal units, the amount is 48 MB.

4. Part B – Network Distribution

The question does not specify the number of distribution events, replicas or protocol overhead, so an exact network traffic value cannot be determined. A simple one-time distribution estimate can be calculated.

29. New certificate per employee = 2 KB.

30. $12,000 \text{ employees} \times 2 \text{ KB} = 24,000 \text{ KB}$.

Therefore, one-time distribution of the new certificates is approximately 24 MB using decimal units.

5. If Both Certificates Are Distributed

31. Old + new = 4 KB per employee.

32. $4 \text{ KB} \times 12,000 = 48,000 \text{ KB}$.

This is approximately 48 MB decimal. Actual traffic can be higher due to network protocol overhead, retries and synchronization.

6. Part C – Lifecycle Challenges

Challenge	Explanation
Certificate expiry	Expired certificates can interrupt secure email.
Renewal management	12,000 renewals are difficult to handle manually.
Revocation	Compromised certificates must be revoked quickly.
Private-key security	Private keys must be protected from unauthorized access.
Old/new overlap	Systems must support the transition correctly.
Employee changes	New devices, transfers and departures require updates.
Trust management	Correct CA and certificate chains must be trusted.

7. Part D – Optimization Strategies

- Automated certificate renewal can renew certificates before expiration.
- A centralized PKI can manage certificate issuance, policy and trust.
- Expiry monitoring can provide alerts before certificates become invalid.
- Automated revocation can respond quickly to compromised certificates.
- Short-lived certificates can reduce the time window of exposure when automated issuance is reliable.
- Directory integration can connect certificates with employee identity information.

8. Best Recommendation

For an organization with 12,000 employees, centralized PKI and automated lifecycle management are the most practical solutions. Manual certificate generation, distribution and renewal should be minimized.

9. Justification

33. There are 12,000 users.
34. Each user needs a certificate.
35. Renewal happens every year.
36. During renewal, old and new certificates overlap.

Therefore, the organization needs automation to track expiry, issue replacements, distribute certificates and revoke invalid certificates. This reduces human error and service interruption.

10. Conclusion

The storage requirement during renewal is 48,000 KB, approximately 46.875 MiB. The main challenge is not storage but managing 12,000 certificate lifecycles reliably. Centralized PKI, automated renewal, monitoring and controlled revocation are the recommended solutions.

QUESTION 5 – Email Encryption Latency in Cloud-Based Systems

The cloud email service uses AES for data encryption and RSA for key exchange. AES takes 0.2 ms per MB, RSA key exchange takes 5 ms per email, each email has an 8 MB attachment, and the system processes 8,000 emails per hour. □filecite□turn1file0□L62-L71□

1. Problem Understanding

- AES is a symmetric encryption algorithm and is efficient for large amounts of data.
- RSA is an asymmetric algorithm and is more computationally expensive.
- The 8 MB attachment is the large data portion of each email.
- The question gives separate times for AES and RSA.
- The goal is to identify the total encryption time and the main bottleneck.

2. Given Data

Parameter	Value
AES time	0.2 ms/MB
RSA key exchange	5 ms/email
Attachment	8 MB
Emails/hour	8,000

3. Part A – Encryption Time Per Email

37. AES time = $0.2 \times 8 = 1.6$ ms.

38. RSA time = 5 ms.

$$\text{Total time/email} = 1.6 + 5 = 6.6 \text{ ms}$$

Therefore, total encryption time per email is 6.6 ms.

4. Part B – Total Processing Time Per Hour

39. Emails/hour = 8,000.

40. Time/email = 6.6 ms.

$$\text{Total} = 8,000 \times 6.6 = 52,800 \text{ ms}$$

41. 52,800 ms = 52.8 seconds.

Therefore, the aggregate cryptographic processing time is 52.8 seconds per hour if all times are simply added. Actual wall-clock time can be lower when cloud processing is parallel.

5. Important Observation

- AES processes 8 MB but takes only 1.6 ms.
- RSA takes 5 ms for each email.
- Therefore, RSA consumes more time even though it is used for key exchange rather than the large data itself.

6. Part C – Bottleneck Calculation

42. Total time = 6.6 ms.

43. RSA time = 5 ms.

$$\text{RSA share} = (5 / 6.6) \times 100 \approx 75.76\%$$

$$\text{AES share} = (1.6 / 6.6) \times 100 \approx 24.24\%$$

Operation	Time/email	Percentage
AES	1.6 ms	24.24%
RSA	5.0 ms	75.76%
Total	6.6 ms	100%

Therefore, RSA key exchange is the main performance bottleneck in this particular scenario.

7. Part D – Optimization Strategies

Strategy	Simple Explanation	Expected Benefit
ECC adoption	Use elliptic-curve public-key cryptography where appropriate.	Can reduce public-key key sizes and improve key-establishment efficiency.
Parallel processing	Process independent emails at the same time.	Improves throughput.
Hardware acceleration	Use optimized cryptographic hardware/CPU features.	Reduces computation time.
Session/key reuse where appropriate	Avoid unnecessary repeated setup operations.	Reduces repeated asymmetric work.
Cloud autoscaling	Add processing resources when workload increases.	Handles high email volume.

8. Best Recommendation

Because RSA accounts for approximately 75.76% of the stated encryption time, the first optimization should target asymmetric key-exchange processing. ECC can be considered where the protocol and security policy support it. Parallel processing is also highly suitable because individual emails can be processed independently.

9. Justification

44. AES = 1.6 ms/email.

45. RSA = 5 ms/email.

46. RSA is more than three times the AES time in this scenario.

Therefore, reducing RSA-related processing will have a greater effect on per-email latency than optimizing AES alone.

10. Conclusion

The total encryption time is 6.6 ms per email, and the aggregate processing time for 8,000 emails is 52.8 seconds. RSA contributes approximately 75.76% of the stated per-email encryption time, making it the main bottleneck. ECC, parallel processing and hardware acceleration are suitable optimization strategies.